

MOEEZ UDDIN

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Final-Year CS Student · Machine Learning · Biomedical Signal Processing · Data Engineering

PROFESSIONAL SUMMARY

Final-year Computer Science student at FAST NUCES with demonstrated expertise in machine learning, EEG/biomedical signal processing, and full-stack database development. Designed and deployed a brain-computer interface system as a capstone project, combining real-time signal acquisition, ML classification, and rigorous validation. Proven ability to deliver across research and client-facing contexts — from academic teaching assistance to freelance data engineering. Seeking roles at the intersection of AI, scientific computing, and interdisciplinary research.

EDUCATION

Bachelor of Science in Computer Science

Aug 2022 – Present

FAST National University of Computing and Emerging Sciences — Peshawar

- **Dean's List Certificate** — Recognised for academic excellence in the second semester.
- Relevant coursework: Data Structures & Algorithms, Operating Systems, Database Systems, Machine Learning, Probability & Statistics, Linear Algebra, Computer Organization & Assembly Language.

Intermediate in Computer Science (Pre-Engineering)

May 2020 – May 2022

Aga Khan Higher Secondary School — Karachi

WORK EXPERIENCE

Teaching Assistant — Mathematics & Computer Science

FAST NUCES | 2023 – Present

- Supported instruction for Calculus, Analytical Geometry, Probability & Statistics, Linear Algebra, and Computer Organization & Assembly Language.
- Conducted lab sessions, graded assessments, and provided one-on-one tutoring to strengthen students' mathematical foundations.

Freelance Developer — Data & ML Projects

2023 – Present

- Delivered end-to-end database systems, data mining pipelines, and interactive analytics dashboards for external clients.
- Developed and fine-tuned Multi-Layer Perceptron (MLP) models for regression and classification tasks on real client datasets.

CAPSTONE PROJECT

Brain-Computer Interface (BCI) for EEG-Based Number Recognition

github.com/Mrup1/final-year-project- | Stack: Python · OpenBCI · scikit-learn · NumPy · Signal Processing

- **Hardware Integration:** Configured the OpenBCI system for multi-channel EEG acquisition; designed and ran controlled experiments to capture stable, high-quality brain signals.
- **Signal Preprocessing:** Implemented noise filtering (band-pass), normalization, and artifact removal pipelines on raw EEG streams to ensure clean classifier input.
- **Feature Engineering & ML:** Extracted time-domain and frequency-domain features from EEG epochs; trained and evaluated multiple classifiers for digit recognition with full cross-validation.
- **Performance Evaluation:** Benchmarked models using accuracy, precision-recall, and validation curves; identified the optimal configuration for real-time deployment.

ACADEMIC & TECHNICAL PROJECTS

Inventory Management System — Database Engineering

- Designed a fully normalised (3NF) MySQL relational schema covering products, suppliers, and transaction workflows.
- Optimised complex SQL queries (indexes, query plans) for large-dataset performance; enforced referential integrity and transactional consistency.

Full-Stack Web Applications — Database & Backend

- Built multiple semester projects using MySQL, PHP, JavaScript, HTML, and CSS with emphasis on backend architecture, schema design, and CRUD operations.
- Implemented role-based access control and dynamic query generation for multi-user web applications.

Operating Systems — Concurrency & Synchronisation

- Implemented a multi-threaded OS-level simulation using POSIX threads and mutexes, demonstrating mastery of synchronisation primitives and deadlock prevention.

Additional Projects

- **Smart Home IoT System** — embedded sensor integration and automated control logic.
- Various system-level and backend projects in C/C++, Java, and scripting environments.

TECHNICAL SKILLS

Programming Languages	Python (pandas, NumPy, scikit-learn, TensorFlow) · SQL · PHP · Java · C# · C/C++ · Google Carbon
Data, AI & Signals	Machine Learning · Deep Learning (MLP/ANN) · NLP · EEG Analysis · Biomedical Signal Processing · Feature Engineering · Power BI · Data Mining · Database Design
Systems & CS Fundamentals	Data Structures & Algorithms · Operating Systems · Multithreading & Concurrency · Computer Architecture
Tools & Platforms	MySQL · Git & GitHub · Linux · OpenBCI · VS Code

LEADERSHIP & ACTIVITIES

- **President, FAST NUCES Sports Society** (Fall 2024 – Fall 2025): Led a student organisation of 100+ members, managing event logistics, budgeting, and inter-university competitions.
- **Technical Team Member, Microsoft Learn Student Ambassador (MLSA) — Peshawar:** Organised technical workshops and supported developer community initiatives.
- **Dean's List Certificate** — Recognised for academic excellence in the second semester.